

# ACTSC 372 Investment Science and Corporate Finance

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## 0 The Role of Financial Intermediaries

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Lecture 1, 2025/09/03

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What is financial investment?

- Ownership / loan contracts helped by financial intermediaries.

### Role of Financial Intermediaries

- **Size intermediation:** the bank gets small lenders together with the “big” borrowers.
- **Term intermediation:** between short term lenders and long term borrowers.
- **Risk intermediation:** pooling resources reduces individual risk.

**Consumer Smoothing:** see handwritten notes.

# 1 Investment Rules

## 1.1 The NPV Rule

**Question:** Given a choice of projects, what project should a rational investor choose?

**Definition 1.1 (Project).**

A **project** is a collection of cash flows  $\{C_0, C_1, C_2, \dots\}$  where  $C_t$  = cash flow at time  $t$ .

*Note.*

- $C_t < 0 \implies$  cost.
- $C_t > 0 \implies$  revenue.

**Definition 1.2 (Pure Investment Project).**

A **pure investment project** is a project where  $C_0 < 0$  and  $C_t \geq 0$  for all  $t > 0$ .

**Definition 1.3 (Pure Financing Project).**

A **pure financing project** is a project where  $C_0 > 0$  and  $C_t \leq 0$  for all  $t > 0$ .

**Definition 1.4 (Independent Projects).**

$A$  and  $B$  are **independent projects** if the cash flows of  $A$  do not affect the cash flows of  $B$ .

*Note.* Between projects  $A$  and  $B$ , we can choose:

- one,
- both,
- or neither.

**Definition 1.5 (Mutually Exclusive Projects).**

$A$  and  $B$  are **mutually exclusive projects** if choosing  $A \implies$  we cannot choose  $B$  and vice versa (we cannot choose both).

**Example.** Given  $C_0, C_1, C_2, \dots$ . How to decide if this is a “good” project?

*Proof.* See NPV rule below. □

**Definition 1.6 (Net Present Value (NPV)).**

The **NPV** of a project is

$$NPV = C_0 + \frac{C_1}{1+r} + \frac{C_2}{(1+r)^2} + \dots = \sum_{t=0}^{\infty} \frac{C_t}{(1+r)^t}$$

where units = dollars.

**Definition 1.7 (NPV Rule).**

- For independent projects, choose all projects with  $NPV > 0$ .
- For mutually exclusive projects, choose the project with the highest  $NPV$ , as long as  $NPV > 0$ .

**Example.** Let  $C_0 = -100, C_1 = 110, C_t = 0$  for  $t \geq 2$ , and  $r = 8\%$ . Then,

$$NPV = -100 + \frac{110}{1.08} > 0.$$

So, we should accept this project.

**Example.** Let  $C_0 = 100, C_1 = -110$ , and  $r = 8\%$ . Then,

$$NPV = 100 - \frac{110}{1.08} < 0.$$

So, we should reject this “bad” project.

What is  $r$ ?  $\implies r$  = “opportunity cost” of the dollar.

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**Lecture 2, 2025/09/08**

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**Advantages of NPV rule:**

- NPV takes into account the time value of money.

- NPV takes into account cash flows, not accounting incomes.

**Example.** Let  $C_0 = 100$ ,  $C_1 = 115$ ,  $r = 10\%$ . The project has positive NPV and the owner will be happy if the project is taken.

**Disadvantages of NPV rule:**

- The cash flows are uncertain.
- What should be the right “ $r$ ”? (see later: CAPM)

## 1.2 Review of ACTSC 231

Some important cash flows to recall from ACTSC 231:

- Propetuity.
- Propetuity with growth.
- Annuity.

**Definition 1.8 (Propetuity, Perpetuity Factor).**

A **propetuity** is an annuity with no end dates (infinite # of payments). Let  $C_t = C$  for all  $t \geq 1$ . Then,

$$\begin{aligned} PV &= \frac{C}{1+r} + \frac{C}{(1+r)^2} + \frac{C}{(1+r)^3} + \cdots \\ &= \frac{C}{1+r} \left[ 1 + \frac{1}{1+r} + \frac{1}{(1+r)^2} + \cdots \right] \\ &= \frac{C}{1+r} \cdot \frac{1}{1 - \frac{1}{1+r}} \\ &= \frac{C}{r}. \end{aligned}$$

The **perpetuity factor** is  $\frac{1}{r}$ .

**Definition 1.9 (Perpetuity with Growth).**

Let  $C_t = C(1 + g)^{t-1}$  for  $t \geq 1$ . Then,

$$\begin{aligned}
 PV &= \frac{C}{1+r} + \frac{C(1+g)}{(1+r)^2} + \frac{C(1+g)^2}{(1+r)^3} + \dots \\
 &= \frac{C}{1+r} \left[ 1 + \frac{1+g}{1+r} + \left( \frac{1+g}{1+r} \right)^2 + \dots \right] \\
 &= \frac{C}{1+r} \cdot \frac{1}{1 - \frac{1+g}{1+r}} \\
 &= \frac{C}{r-g}, \quad g < r.
 \end{aligned}$$

**Definition 1.10 (Annuity, Annuity Factor).**

Let  $C_t = C$  for  $1 \leq t \leq n$ . Then,

$$\begin{aligned}
 PV &= \frac{C}{1+r} + \frac{C}{(1+r)^2} + \dots + \frac{C}{(1+r)^n} \\
 &= \frac{C}{1+r} \left[ 1 + \frac{1}{1+r} + \dots + \frac{1}{(1+r)^{n-1}} \right] \\
 &= \frac{C}{1+r} \cdot \frac{1 - \left( \frac{1}{1+r} \right)^n}{1 - \frac{1}{1+r}} \\
 &= C \cdot \frac{1 - (1+r)^{-n}}{r}.
 \end{aligned}$$

The **annuity factor** is  $\frac{1}{r} \left[ 1 - \left( \frac{1}{1+r} \right)^n \right]$ .

**1.3 Other Investment Rules****Definition 1.11 (Payback Period).**

The **payback period** is the # of years required to recoup the initial investment (ignoring the time value of money).

**Example.** Let  $C_0 = -300$ ,  $C_t = 100$  for  $t \geq 1$ . The payback period is 3 years.

*Note.* The units of payback period is years.

**Definition 1.12 (Payback Period Rule).**

Decide on a cutoff period,  $T^*$ . If the payback period  $\leq T^*$ , accept; otherwise, reject it.

**Disadvantages of Payback Period Rule:**

- It does not take into account the time value of money.
- The cutoff is arbitrary.
- Negative NPV may be accepted.
- Positive NPV may be rejected.

**Definition 1.13 (Discounted Payback Period).**

The **discounted payback period** is the payback period, but without ignoring the time value of money.

**Example.** Let  $C_0 = -300$ ,  $C_t = 100$  for  $t \geq 1$ , and  $r = 12.5\%$ . We are solving  $\sum_{t=1}^T \frac{C_t}{(1+r)^t} = -C_0$ . This implies that the discounted payback period,  $T$ , is 4 years.

**Definition 1.14 (Internal Rate of Return (IRR)).**

The **internal rate of return (IRR)** of a project is the cost of capital such that the NPV is zero.

**Example.** Let  $C_0 = -100$ ,  $C_1 = 112$ ,  $r = 10\%$ . We have two ways to determine if we should accept this project:

- (1) Calculate NPV:  $NPV = -100 + \frac{112}{1.1} > 0 \implies$  accept.
- (2) The project has a return of 12%. Since it is bigger than  $r = 10\%$ , we should accept it.

**Question:** How to solve for the IRR?

**Answer:** Calculate the NPV and equal it to zero, then solve for  $r^*$ . That is,

$$\sum_{t=0}^T \frac{C_t}{(1+r^*)^t} = 0.$$

**Definition 1.15 (IRR Rule).**

Calculate the IRR of a project.

- For independent projects, choose all projects with  $IRR > r$ .
- For mutually exclusive projects, choose the project with the highest IRR (as long as  $IRR > r$ ).

**Advantages of IRR rule:**

- In order to calculate IRR, we do not need  $r$ .
- It usually agrees with the NPV rule.

**Disadvantages of IRR rule:**

The first two disadvantages are for independent projects.

- (1) IRR cannot distinguish between investment and financing.

*Note.*  $(-1) \sum \frac{C_t}{(1+r^*)^t} = 0(-1)$ , the solution remains the same.

**Example.** Let  $C_0 = -100, C_1 = 112, r = 10\%$ . The IRR is 12% and we should accept this project.

Let  $C_0 = 100, C_1 = -112, r = 10\%$ . The IRR is still 12% but we should reject this project.

- (2) Polynomial equation problem: finding the IRR = solving a polynomial equation, i.e.

$$C_0 + \frac{C_1}{1+r^*} + \frac{C_2}{(1+r^*)^2} + \cdots + \frac{C_T}{(1+r^*)^T} = 0.$$

However, we could have no roots, multiple roots, imaginary roots, or complex roots.

**Example.** Let  $C_0 = 0, C_1 = 100, C_t = 0$  for all  $t \geq 2$ . The  $IRR = \infty$ .

**Example.** Let  $C_0 = -100, C_1 = 230, C_2 = -132$ . Then,

$$-100 + \frac{230}{1+r^*} - \frac{132}{(1+r^*)^2} = 0 \implies r^* = 10\% \text{ or } 20\%.$$

This is problematic, since which IRR should we use?



We will need the following result to deal with the multiple IRR problem.

**Theorem 1.1 (Descartes' Rule of Signs).**

The # of positive roots of a polynomial is  $\leq$  # of “sign switches” of the coefficients.

**Example.** For the previous example where  $C_0 = -100$ ,  $C_1 = 230$ ,  $C_2 = -132$ , there are 2 sign switches. Thus, there are at most 2 positive roots.

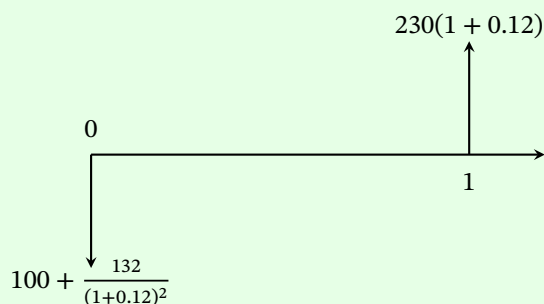
We have the following MIRR to avoid multiple solutions.

**Definition 1.16 (Modified Internal Rate of Return (MIRR)).**

The **modified IRR**

- brings all negative cash flows to time  $t = 0$ , and
- brings all positive cash flows to time  $t = T$ ,

**Example.** Let  $C_0 = -100$ ,  $C_1 = 230$ ,  $C_2 = -132$ , and  $r = 12\%$ . Then,



Calculate the IRR of the modified project (MIRR):

$$-\left(100 + \frac{132}{(1 + 0.12)^2}\right) + \frac{230(1 + 0.12)}{1 + r^*} = 0 \implies r^* = 12.36\%.$$

*Note.* We only have 1 “sign switch” in the modified project, so MIRR is unique. Then, we can compare MIRR with  $r$ , and choose the project with  $\text{MIRR} > r$ . But we need to know today’s interest rate  $r$ .

**Disadvantages of IRR rule (continued):**

For mutually exclusive projects (i.e. we can only choose one project.):

(3) Scaling problem: the IRR is biased towards “smaller” projects.

**Example.**

Project A:  $C_0 = -1, C_1 = 2$ .

Project B:  $C_0 = -100, C_1 = 150$ .

Note that  $IRR_A = 100\%$  and  $IRR_B = 50\%$ . However, project B is a better one.

Note.  $\sum \frac{kC_t}{(1+r)^t} = 0$ . IRR remains the same if you multiply by a constant.

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**Lecture 3, 2025/09/10**

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Here are some ways to deal with the scaling problem:

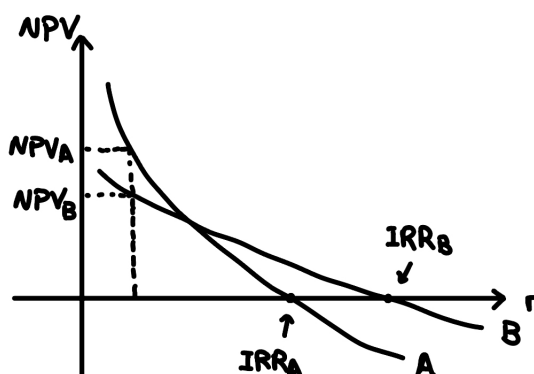
- Calculate the NPV and choose the higher one. In the example,  $IRR_A > IRR_B$ , but  $NPV_A = 1 < NPV_B$ .
- If the IRR picks the “smaller” project, then we do **incremental analysis**.

**Example.** Incremental project:  $C = B - A$  with  $C_0 = -99, C_1 = 148$ .

If the IRR of the incremental project  $< r \Rightarrow$  stick with the smaller project. If  $> r \Rightarrow$  go with the bigger project.

(4) Timing problem.

Consider pure investment projects.



According to the IRR rule, B would be chosen, but A is better because  $NPV_A > NPV_B$ .

**Example.**

Project short:  $C_0 = -300, C_1 = 345, C_t = 0$  for  $t \geq 2$ .

Project long:  $C_0 = -200, C_4 = 330, C_t = 0$  for  $t \neq 0, 4$ .

Note that

$$IRR_{\text{short}} = 15\%, \quad NPV_{\text{short}} = 13.64$$

$$IRR_{\text{long}} = 13.34\%, \quad NPV_{\text{long}} = 25.39.$$

*Note (Recall from ACTSC 231: YTM of a Bond).*

A bond is a loan in exchange for regular coupon payments and the redemption value at maturity.

We have

$$P = \frac{C}{1+r} + \frac{C}{(1+r)^2} + \cdots + \frac{C+F}{(1+r)^N}$$
$$0 = -P + \frac{C}{1+r} + \frac{C}{(1+r)^2} + \cdots + \frac{C+F}{(1+r)^N}.$$

Note that  $r$  is the interest rate that equates NPV to 0. So YTM = IRR of the bond.

**Example.** Let  $C_0 = -100, C_1 = 10, C_2 = 110$  with  $r = 10\%$ . We can think of this as a bond with  $P = 100, C = 10, F = 110, N = 2$ . The IRR is 10%.

Also note that if the money was put in a bank, then we would have  $C_0 = -100, C_2 = 121$ .

The extra \$1 is due to the \$10 invested at 10% for another year.

IRR assumes that all cash flows can be reinvested at the IRR itself.

A more realistic assumption that cash flows can be reinvested at  $r$ , not IRR.

**Definition 1.17 (Profitability Index (PI)).**

The **profitability index** is

$$PI = \frac{\text{PV of future cash flows}}{\text{Initial Investment}} = \frac{\sum_{t=1}^{\infty} \frac{C_t}{(1+r)^t}}{-C_0}.$$

**Definition 1.18 (PI Rule).**

- For independent projects, choose all projects with  $PI > 1$ .
- For mutually exclusive projects, choose the project with the highest  $PI$  with  $PI > 1$ .

**Proposition 1.2.** For independent projects, PI agrees with the NPV rule.

*Proof.* Note that

$$\frac{\sum_{t=1}^{\infty} \frac{C_t}{(1+r)^t}}{-C_0} > 1 \implies NPV > 0.$$

□

**Disadvantages of PI rule:**

For mutually exclusive projects:

- (1) Scaling problem (same as the IRR rule).

**Example.** Let  $r = 10\%$ .

Project A:  $C_0 = -10, C_1 = 15, C_2 = 40$ .

Project B:  $C_0 = -20, C_1 = 70, C_2 = 10$ .

Note that

$$PI_A = 4.53, \quad NPV_A = 35.3$$

$$PI_B = 3.52, \quad NPV_B = 50.5.$$

Note that  $PI_A > PI_B$ , but  $NPV_A < NPV_B$ . Let's consider the incremental project  $C = B - A$  with  $C_0 = -10, C_1 = 55, C_2 = -30$ .

If the incremental project has  $PI > 1$ , then choose the bigger project; otherwise, choose the smaller project.

**Summary of Investment Rules:** NPV rule is the best.

## 1.4 Equivalent Annual Costs (EAC)

### Example (Machines with Unequal Lives).

Let  $r = 10\%$  and consider the following two machines.

A: \$500 to buy, \$120/year to operate, lasts for 3 years.

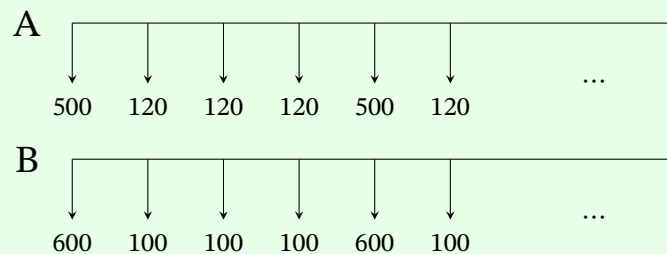
B: \$600 to buy, \$100/year to operate, lasts for 4 years.

Note that we have to use the machine (so we buy another one after it breaks down).

We can calculate the NPVs to get

$$NPV_A = -798.42, \quad NPV_B = -916.94.$$

However, the actual cash flows are different. We have the following cash flows:



Find the EAC of each machine.

*Proof.* This is equivalent to solving for  $x$  in  $NPV_A = x \cdot a_{\overline{3}|10\%}$  AND  $NPV_B = x \cdot a_{\overline{4}|10\%}$ . Then, we get

$$EAC_A = \frac{NPV_A}{a_{\overline{3}|10\%}}, \quad EAC_B = \frac{NPV_B}{a_{\overline{4}|10\%}}.$$

We choose the one with lower EAC! □

**Goal:** Calculate NPV, cash flows,  $r$ , in the right way!

## 2 Capital Budgeting

### 2.1 Rules for Calculating NPVs

Remember the following rules when calculating NPVs:

- (1) Ignore sunk costs, i.e. costs we have to incur with/without the project.

**Example.** Thaler Pizza experiment.

- (2) Include opportunity costs, i.e. cost of the next best alternative.

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#### Lecture 4, 2025/09/15

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- (3) Be aware of externalities, i.e. effects on other projects. One project can affect cash flows from other projects.

**Example.**

- Positive externality: synergy.
- Negative externality: erosion.

- (4) Be aware of the difference between “real” and “nominal”.

*Note.*

- Nominal Interest ( $i$ ): in dollars.
- Real interest rate ( $r$ ): amount of extra goods you can buy.
- Rate of inflation ( $\pi$ ): increase with prices.

*Example.* Suppose you have \$1 today and price of the good is \$1. If you put \$1 in the bank, it will become  $1 + i$  dollars in one year. Price of the good in one year will be  $1 + \pi$ . So the # of goods you can buy next year is

$$\frac{1 + i}{1 + \pi} = 1 + r.$$

The # of extra units of the good you can buy is (Fisher's Law):

$$r = \frac{1+i}{1+\pi} - 1 = \frac{i-\pi}{1+\pi}.$$

We can also approximate  $r$  by

$$r \approx i - \pi.$$

### Rule for Calculating Cash Flows

- Be consistent. If we are using real cash flows, use real interest rates, otherwise use nominal cash flows and nominal interest rates.

*Note.* Nominal cash flows to real cash flows:

Divide by  $(1 + \pi)^t$ .

Real cash flows to nominal cash flows:

Multiply by  $(1 + \pi)^t$ .

**Example.** Suppose all cash flows are nominal. Let  $C_0 = -1000$ ,  $C_1 = 600$ ,  $C_2 = 650$ ,  $i = 14\%$ ,  $\pi = 5\%$ . Calculate the NPV.

*Proof.*

**Method 1:** Use nominal.

$$NPV = -1000 + \frac{600}{1 + 0.14} + \frac{650}{(1 + 0.14)^2}.$$

**Method 2:** Find the real interest rate from Fisher's Law:

$$1 + r = \frac{1 + i}{1 + \pi}.$$

Let  $C'_0$  be the real cash flow at time 0, then

$$\begin{aligned}C'_0 &= C_0 \\C'_1 &= \frac{C_1}{1 + \pi} \\C'_2 &= \frac{C_2}{(1 + \pi)^2}.\end{aligned}$$

Then,

$$NPV = C'_0 + \frac{C'_1}{1 + r} + \frac{C'_2}{(1 + r)^2}.$$

We should get the same answer using both methods. □

*Remark (Why does this work?).*

Suppose all cash flows are nominal. Then,

$$\begin{aligned}NPV &= C_0 + \frac{C_1}{1 + i} + \frac{C_2}{(1 + i)^2} + \dots \\&= C_0 + \frac{\frac{C_1}{1 + \pi}}{\frac{1 + \pi}{1 + i}} + \frac{\frac{C_2}{(1 + \pi)^2}}{\left(\frac{1 + i}{1 + \pi}\right)^2} + \dots \\&= C'_0 + \frac{C'_1}{1 + r} + \frac{C'_2}{(1 + r)^2} + \dots.\end{aligned}$$

- (5) Be aware of depreciation for cash flows from accounting statements.

**Example.**

| Revenue | Cost | Depreciation | Taxes |
|---------|------|--------------|-------|
| 1500    | 700  | 600          | 34%   |

Depreciation is a tax-deductible expense! For the taxable income,

$$\begin{aligned}\text{Taxable Income} &= \text{Revenue} - \text{Cost} - \text{Depreciation} \\&= 1500 - 700 - 600 = 200.\end{aligned}$$

Also,

$$\text{Tax paid} = 0.34 \times 200 = 68.$$



Then,

$$\text{Cash Flow} = \text{Revenue} - \text{Cost} - \text{Tax paid} = 1500 - 700 - 68 = 732.$$

General approach: Let the cash flow be  $CF_t$ . Then,

$$\begin{aligned} CF_t &= \text{Revenues} - \text{Costs} - \text{Taxes} \\ &= (R_t - \text{Cost}_t) - (R_t - \text{Cost}_t - D)T_c \\ &= (R_t - \text{Cost}_t)(1 - T_c) + DT_c. \end{aligned}$$

The term  $(R_t - \text{Cost}_t)(1 - T_c)$  is called the **after tax operating cash flow (ATOCF)**.

Note that the term  $DT_c$  is the tax break we get because we can subtract depreciation.

And  $DT_c$  is the **capital cost allowance tax shield (CCATS)**.

$$\begin{aligned} NPV &= C_0 + \frac{C_1}{1+r} + \dots + \frac{C_t}{(1+r)^t} \\ &= C_0 + \frac{C_1}{1+r} + \dots + \frac{(\text{ATOCF} + \text{CCATS})}{(1+r)^t} \\ &= C_0 + PV\text{ATOCF} + PV\text{CCATS}. \end{aligned}$$

Note that  $PV(\text{ATOCF})$  is the PV without subtracting depreciation, and  $PV(\text{CCATS})$  is the PV of all tax breaks.

*Note.* To see the above, we have

$$\begin{aligned} NPV &= C_0 + \frac{C_1}{1+r} + \dots + \frac{C_t}{(1+r)^t} + \dots \\ &= C_0 + \frac{(\text{ATOCF})_1 + (\text{CCATS})_1}{1+r} + \dots + \frac{(\text{ATOCF})_t + (\text{CCATS})_t}{(1+r)^t} + \dots \\ &= C_0 + \sum_{t=1}^{\infty} \frac{(\text{ATOCF})_t}{(1+r)^t} + \sum_{t=1}^{\infty} \frac{(\text{CCATS})_t}{(1+r)^t}. \end{aligned}$$

**Question:** How is depreciation ( $D$ ) calculated?

**Example.** Suppose  $d = 20\%$ . A machine costs \$1 million dollars.

| Year | UCC Beginning | Depreciation ( $D$ ) | UCC End   |
|------|---------------|----------------------|-----------|
| 1    | \$1,000,000   | \$200,000            | \$800,000 |
| 2    | \$800,000     | \$160,000            | \$640,000 |
| 3    | \$640,000     | \$128,000            | \$512,000 |

Note that  $D = d \cdot \text{UCC Beginning}$ . UCC is the **undepreciated capital cost**.

**Half-Year Rule:** if you buy an asset, only 50% of the purchase price can be added to the first year. The remaining 50% can be added in the second year.

**Example (Half-Year Rule).**

Suppose  $d = 20\%$  and a machine costs \$1 million dollars.

| Year | UCC Beginning | Depreciation ( $D$ ) | UCC End   |
|------|---------------|----------------------|-----------|
| 1    | \$500,000     | \$100,000            | \$400,000 |
| 2    | \$900,000     | \$180,000            | \$720,000 |
| 3    | \$320,000     | \$144,000            | \$576,000 |

**Theorem 2.1 (PVCCATS Formula).**

We have

$$PV(\text{CCATS}) = \frac{C \cdot d \cdot T_c}{r + d} \cdot \frac{1 + 0.5r}{1 + r} - \frac{S \cdot d \cdot T_c}{r + d} \cdot \frac{1}{(1 + r)^n}$$

where

- $C$  = cost of the asset.
- $d$  = rate of depreciation.
- $n$  = lifetime of the asset.
- $S$  = salvage value of the asset.

## 2.2 Capital Budgeting Under Uncertainty

**Example.** MATHSOC is thinking of installing a solar panel:

- Cost: \$1 million.
- Annual operating cost: 10K.
- Lifetime: 15 years.
- Salvage: 100K.
- Revenues: \$100/hour.
- # of sunny hours: 1500.
- $r = 10\%$ .
- No taxes.

Find the NPV.

*Proof.* We have

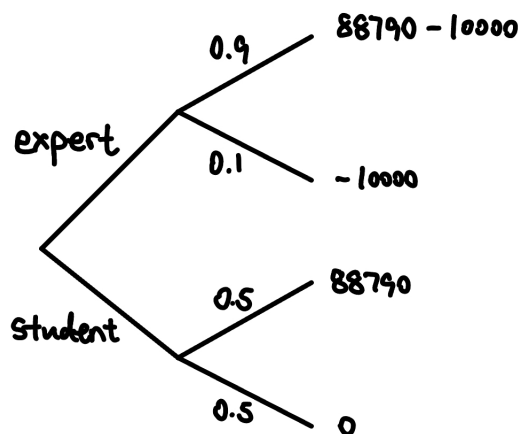
$$NPV = -1000000 - 10000a_{\overline{15}|10\%} + 150000a_{\overline{15}|10\%} + \frac{100000}{(1.1)^{15}} = 88790.$$

□

### Decision Tree

For the MATHSOC example, installation may not work. Consider two options:

- Hire an expert: cost \$10000, success rate 90%.
- Hire a student: cost \$0, success rate 50%.



There are two types of nodes in a decision tree:

- Probability nodes: calculate expected values.
- Decision nodes: choose the “better” path.

**Example (Continued).**

At the expert node, the expected value is  $0.9(88790) + 0.1(-10000) = 76911$ .

At the student node, the expected value is  $0.5(88790) + 0.5(-10000) = 39395$ .

Since  $76911 > 39395$ , we should hire the expert.

**Scenario Analysis**

For the MATHSOC example, what if the # of sunny hours  $\neq 1500$ ? Suppose # of sunny hours are 1100, 1500, 1900 with probability  $\frac{1}{3}$  each.

We calculate the expected NPV. ????

**Monte Carlo Simulation**

Suppose sunny hours  $\sim N(1500, 200)$ . We simulate  $n$  observations from this distribution, and calculate NPVs for each outcome, and look at the distribution of NPVs.

**Sensitivity Analysis**

How sensitive is our NPV to different variables? Let  $S$  = # of sunny hours. Then, we want to find  $\frac{dNPV}{dS} \implies$  how accurate we need to be with our estimates.

**Break-Even Analysis**

Break-even point is the value of the variable that makes  $NPV = 0$ .

**Definition 2.1 (Contribution Margin).**

The **contribution margin** is the increase in net income per unit increase in sales, i.e.

$$\text{Contribution Margin} = (\text{Unit Revenue} - \text{Unit Cost}) \cdot (1 - T_c)$$

where  $T_c$  is the corporate tax rate.

**Example.** The Tim Hortons at SLC sells donuts. Assume no taxes.

- Donuts: \$1.20.
- Cost of donuts: \$0.70.
- Fixed cost: \$100,000.

**Contribution Margin** =  $1.20 - 0.70 = 0.50$ .

Then, the break-even point for # of donuts sold is  $\frac{100000}{0.50} = 200000$ .

**Example.** Same setting as above example, but with tax = 40%. Then,

$$\text{Contribution Margin} = (1.20 - 0.70)(1 - 0.4) = 0.30.$$

Then, the break-even point for # of donuts sold is  $\frac{100000}{0.30} \approx 333333$ .

**Example.** The IRR is the break-even interest rate!

**Example.** The discounted payback period is the break-even time!

For the next few lectures, we will focus on the following question.

**Question:** What is the right “ $r$ ” for a project? (CAPM)

## 2.3 Utility Theory

Before we get there, let’s consider the following question.

**Question:** How do rational human beings evaluate uncertain outcomes?

**Definition 2.2 (Lottery).**

A lottery  $X$  is a random variable whose outcomes are dollars.

**Example.** Consider the lotteries  $X$  and  $Y$  with the following PMFs:

|                     |     |     |
|---------------------|-----|-----|
| $X$                 | 100 | 25  |
| $\mathbb{P}(X = x)$ | 0.5 | 0.5 |

|                     |     |     |
|---------------------|-----|-----|
| $Y$                 | 81  | 49  |
| $\mathbb{P}(Y = y)$ | 0.5 | 0.5 |

One way to compare lotteries is to compare their expected values.

**Example (ST. Petersburg Paradox).**

Let  $X$  be the amount of money you win from the following lottery. A fair coin is tossed until it comes up heads. If the first head appears on the  $n$ -th toss, you win  $2^n$  dollars.

| $X$      | Probability    |
|----------|----------------|
| 2        | $\frac{1}{2}$  |
| 4        | $\frac{1}{4}$  |
| 8        | $\frac{1}{8}$  |
| 16       | $\frac{1}{16}$ |
| $\vdots$ | $\vdots$       |

The expected value of the lottery is

$$\mathbb{E}[X] = \sum x_i p_i = \frac{1}{2} \cdot 2 + \frac{1}{4} \cdot 4 + \frac{1}{8} \cdot 8 + \dots = 1 + 1 + 1 + \dots = \infty.$$

So you should be willing to pay any finite amount to play this game. However, in reality, most people are not willing to pay a lot of money to play this game.

**John-von-Neumann: Expected Utility Theory**

We use the preference relation  $\succeq$  to denote “is at least as good as”. That is,

$$x \succeq y \implies x \text{ is at least as good as } y.$$

**Question:** What assumptions should  $\succeq$  follow for rational agents?

**Assumptions**

- (1) Completeness:  $\forall X, Y$ , either  $X \succeq Y$  or  $Y \succeq X$  (or both).
- (2) Transitivity:  $\forall X, Y, Z$ , if  $X \succeq Y$  and  $Y \succeq Z$ , then  $X \succeq Z$ .
- (3) Independence:  $\forall X, Y, Z$  and  $\forall p \in (0, 1)$ ,  $X \succeq Y \iff pX + (1 - p)Z \succeq pY + (1 - p)Z$ .

**Theorem 2.2.** If  $\succeq$  satisfies the above three assumptions, then  $\exists$  a function  $u(\cdot)$  such that  $\forall X, Y$ ,

$$X \succeq Y \iff \mathbb{E}[u(X)] \geq \mathbb{E}[u(Y)].$$

Note.  $u(\cdot)$  is called the **von-Neumann Morgenstern utility function**. It is the utility you get after consuming  $x$  dollars.

**Example.** Consider the following lotteries with  $u(x) = \sqrt{x}$ .

|                     |     |     |                     |     |     |
|---------------------|-----|-----|---------------------|-----|-----|
| $X$                 | 100 | 25  | $Y$                 | 81  | 49  |
| $\mathbb{P}(X = x)$ | 0.5 | 0.5 | $\mathbb{P}(Y = y)$ | 0.5 | 0.5 |

For lottery  $X$ ,

$$\mathbb{E}[u(X)] = 0.5\sqrt{100} + 0.5\sqrt{25} = 5 + 2.5 = 7.5.$$

For lottery  $Y$ ,

$$\mathbb{E}[u(Y)] = 0.5\sqrt{81} + 0.5\sqrt{49} = 4.5 + 3.5 = 8.$$

## 2.4 Risk Aversion

**Definition 2.3 (Risk Averse).**

A person is a **risk averse** if (for any lottery), they prefer the expected values for sure over the lottery itself.

**Definition 2.4 (Risk Neutral).**

A person is **risk neutral** if they only care about the expected values.

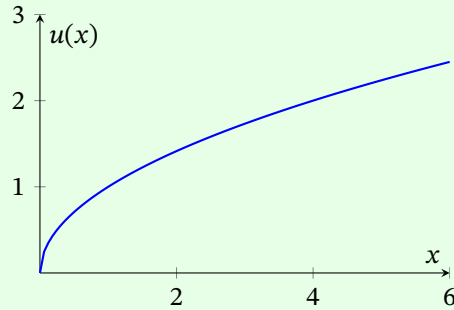
**Definition 2.5 (Risk Loving).**

A person is **risk loving** if they prefer the lottery over the expected values.

**Proposition 2.3.** If the von-Neumann Morgenstern utility function  $u(\cdot)$

- is concave, then the person is risk averse.
- is linear, then the person is risk neutral.
- is convex, then the person is risk loving.

**Example.** Let  $u(x) = \sqrt{x}$ . Note that  $u''(x) < 0$ , so the person is risk averse.



Decreasing marginal utility over money  $\implies$  Risk averse, because the more money you have, the less utility you get from an extra dollar.

Say we have lotteries  $X$  and  $Y$  with the following PMFs:

|                     |     |     |
|---------------------|-----|-----|
| $X$                 | 100 | 0   |
| $\mathbb{P}(X = x)$ | 0.5 | 0.5 |

|                     |    |
|---------------------|----|
| $Y$                 | 50 |
| $\mathbb{P}(Y = y)$ | 1  |

Then,

$$\mathbb{E}[u(X)] = 0.5\sqrt{100} + 0.5\sqrt{0} = 5$$

$$\mathbb{E}[u(Y)] = 1 \cdot \sqrt{50} \approx 7.07.$$

If  $u(x) = x$ , then  $\mathbb{E}[u(X)] = 0.5 \cdot 100 + 0.5 \cdot 0 = 50$  and  $\mathbb{E}[u(Y)] = 1 \cdot 50 = 50$ . So the person is risk neutral.

If  $u(x) = x^2$ , then  $\mathbb{E}[u(X)] = 0.5 \cdot 100^2 + 0.5 \cdot 0^2 = 5000$  and  $\mathbb{E}[u(Y)] = 1 \cdot 50^2 = 2500$ . So the person is risk loving.



## A Tutorials

### A.1 Tutorial 1

#### Mobius 2 Hints

Q1: Ignore CCA = cannot subtract depreciation.

Q10:  $\sigma(NPV)$  is the standard deviation of NPV.

*Note.* For  $Y = a + bX$ ,  $\sigma_Y = |b|\sigma_X$ .

#### Expected Utility

$U(X)$  = utility you get from lottery  $X$ . Under certain assumptions,

$$U(X) = \mathbb{E}[u(X)] = \sum u(x)f(x)$$

where  $u(x)$  is the von-Neumann Morgenstern utility function and  $f(x)$  is the PMF of  $X$ .

#### Risk Aversion

- A person is risk averse if  $u(\mathbb{E}[X]) > \mathbb{E}[u(X)]$  for all  $X$ .
- A person is risk neutral if  $u(\mathbb{E}[X]) = \mathbb{E}[u(X)]$  for all  $X$ .
- A person is risk loving if  $u(\mathbb{E}[X]) < \mathbb{E}[u(X)]$  for all  $X$ .

*Note.*  $u(\cdot)$  is a concave function  $\iff$  risk averse.

**Example.** Let  $X$  be a lottery with PMF  $\mathbb{P}(X = 64) = \mathbb{P}(X = 8) = 0.5$  and let  $Y$  be a lottery with PMF  $\mathbb{P}(Y = 125) = \mathbb{P}(Y = 0) = 0.5$ . Consider  $u(x) = x^{1/3}$  and  $u(x) = \sqrt{x}$ .

Note that for  $X$  with  $u(x) = x^{1/3}$ ,

$$U(X) = 0.5 \cdot 64^{1/3} + 0.5 \cdot 8^{1/3} = 3.$$

For  $Y$  with  $u(x) = x^{1/3}$ ,

$$U(Y) = 0.5 \cdot 125^{1/3} = 2.5.$$

Find  $U(X)$  and  $U(Y)$  for  $u(x) = \sqrt{x}$ .

## Certainty Equivalent

Let  $CE[X]$  = amount of dollars for sure that makes you indifferent with the lottery  $X$ .

- Risk averse:  $CE[X] < E[X]$ .
- Risk neutral:  $CE[X] = E[X]$ .
- Risk loving:  $CE[X] > E[X]$ .

### Definition A.1 (Certainty Equivalent).

The **certainty equivalent** of a lottery  $X$  is the amount of money  $CE[X]$  such that

$$u(CE[X]) = E[u(X)].$$

If  $u(\cdot)$  is strictly increasing,  $CE[X] = u^{-1}(E[u(X)])$ .

**Example.** Consider the following:

- Wealth: \$100.
- Fine: \$36 with probability  $p = \frac{1}{4}$ .
- M: \$9.25, parking fee to avoid the fine.
- Utility function:  $u(x) = \sqrt{x}$ .

*Proof.* Pay for parking =  $\sqrt{100 - M}$ . Don't pay for parking =  $\frac{1}{4}\sqrt{100 - 36} + \frac{3}{4}\sqrt{100} = 9.5$ .

So  $\sqrt{100 - M} > 9.5 \implies M < 100 - 9.5^2 = 9.75$ . So we should pay for parking.  $\square$

## Are We Rational?

### Example (Allais Paradox).

Consider the following choices:

- A: \$1 million dollars with  $p = 1$ .
- B: \$1 million dollars with  $p = 0.89$ , \$5 million dollars with  $p = 0.1$ , and \$0 with  $p = 0.01$ .
- C: \$0 with  $p = 0.9$ , \$1 million dollars with  $p = 0.1$ .
- D: \$0 with  $p = 0.89$ , \$1 million dollars with  $p = 0.11$ .

If you prefer A over B and C over D, you are not rational.